



Rayat Shikshan Sanstha's
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Accredited With 'B++' Grade By 'NAAC'



Department of Computer Science and Engineering

A Project Design Report Phase I

Gold Loan Management Application

(FinTech)

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Academic Year 2025–26

Certificate

This is to certify that the part of the work of the project entitled “**Gold Loan Management Application (FinTech)**” is submitted by:

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Thanking You,

1. Prathmesh Kailas Jadhav
2. Pranav Suhas Ghorpade
3. Ajay Laxman Madane

Abstract

The Gold Loan Management Application represents a digital solution engineered to streamline and contemporize the management of gold loans within jewelry establishments. Conventional practices, including the utilization of handwritten ledgers, manual interest calculations, and the tracking of repayment schedules, are susceptible to inaccuracies, are labor-intensive, and become increasingly unwieldy as business operations expand. This initiative seeks to mitigate these difficulties through the creation of a mobile application, employing **Flutter** for the frontend and **Spring Boot, Spring Data JPA**, and a **SQL** database for the backend infrastructure. Through a structured workflow, the system automates important tasks like customer registration, gold loan creation, interest and EMI calculations, repayment tracking, and automated due-date reminders. A clear and user-friendly user interface guarantees usability for non-technical staff, while the backend offers accurate financial computations and secure and dependable data storage. The application also includes a dashboard for real-time loan monitoring and summary reports, allowing jewelry store owners to make informed decisions.

Performance analysis demonstrates the application's accuracy, scalability, and efficiency for practical use. In terms of accuracy, operational effectiveness, reminders, and data security, comparative analysis also shows notable improvements over conventional manual methods. All things considered, the Gold Loan Management Application effectively offers a cutting-edge, dependable, and user-friendly platform that boosts productivity and facilitates the digital transformation of gold loan services in jewelry stores.

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Chapter 1

Introduction

One of the most popular secured lending options in India is gold loans, particularly for small and medium-sized jewelry stores. Customers use this model to pledge their gold ornaments as collateral and get paid according to the gold's appraised value. The customer can then retrieve their jewelry once the loan has been paid back with interest within the predetermined time frame.

Numerous jewelry stores continue to operate using manual systems such as notebooks, physical files, or basic spreadsheets, despite the financial importance of gold loans. Tracking due dates, accurately calculating interest, maintaining historical records, and ensuring timely reminders for repayments are difficult with these traditional methods. As customer volume increases, manual handling becomes time-consuming, prone to errors, and inefficient.

By offering a digital platform created especially for jewelry stores, the **Gold Loan Management Application** is intended to address these issues. Shop owners and employees can use the system to create and monitor gold loans, manage customer records, compute interest and EMIs, and automatically and systematically generate repayment reminders.

The application is made up of a **Spring Boot-based backend** that is integrated with **Spring Data JPA** and a **SQL** database to provide safe and dependable data storage, and a **Flutter**-based frontend that provides an easy-to-use mobile interface. When combined, these technologies produce a reliable, scalable, and effective system that improves overall management and simplifies daily loan operations.

Chapter 2

Literature Review

Digital loan management has become an important area of research in financial technology, particularly for gold-backed lending. Research shows that manual systems used by small institutions are often inefficient and prone to errors.

1. **International Journal of Financial Technology and Innovations, Sharma & Verma (2021):** It was discovered that small institutions continue to process gold loans using manual, paper-based registers. They highlighted missing records and frequent valuation errors. The study concluded that digital automation enhances accuracy, record preservation, and loan tracking, directly supporting the goal of this project.
2. **Journal of Banking and Information Systems, Kumar & Singh (2020):** This study examined automated systems for loan documentation. It claimed that digital interest and EMI calculators significantly reduce human error. Their findings validate the automation modules and calculator used in the Gold Loan Management Application.
3. **International Journal of Advanced Research in Computer Science, Patel & Shah (2020):** The authors contrasted manual and digital loan workflows. They observed issues such as incomplete records, slow data retrieval, and inaccurate calculations in manual processes. Their research strongly supports the use of SQL databases and ORM frameworks such as Spring Data JPA.
4. **International Journal of Computer Science and Mobile Computing, Rodrigues & Fernandes (2020):** This study demonstrated that Flutter provides quick development cycles and a smooth, cross-platform user interface. It emphasized that Flutter is well-suited for mobile-first financial applications, supporting the frontend technology of this project.
5. **International Journal of Web Engineering Technologies, Rahman & Abdullah (2021):** The authors discussed RESTful architectures built using Spring Boot. They showed that layered architectures improve backend security, maintainability, and reliability. Their conclusions align with the backend design of this application.

6. **Thomas & Joseph, IEEE Publication (2021):** This research investigated automated loan systems for small businesses. It concluded that structured workflows and system-generated audit logs improve repayment transparency and discipline. Their findings support the use of automated reminders and audit-friendly architecture in this project.

Chapter 3

Objectives

3.1 Problem Statement

The underlying management procedure in many jewelry stores is still unstructured and informal, despite the widespread use of gold loan services. This results in a number of specific issues that can be expressed formally as follows:

- Unorganized and manual record-keeping
- Interest and Repayment Calculations That Are Prone to Errors
- Inadequate customer loan history
- Limited visibility and tracking of active loans
- Absence of automated loan reminders
- Data security and backup issues

Problem Statement:

”How to create a digital, automated, and user-friendly gold loan management system that streamlines loan processing in jewelry stores, guarantees precise computations, sends out timely reminders, keeps safe records, and enhances overall operational efficiency?”

3.2 Project Objectives

The Gold Loan Management Application’s main objective is to convert the conventional, manual gold loan procedure into a dependable, automated, and structured digital system appropriate for jewelry stores. The application was created with the following particular goals in order to accomplish this:

- To enable jewelry stores to keep comprehensive customer, loan, and repayment records in a digital format, thereby digitizing loan management procedures.
- To ensure precise and error-free results by automating financial computations like interest, EMI, and outstanding amounts.
- To improve customer communication and repayment tracking by offering an automated reminder system that notifies store owners about loans that are past due.
- To create a simple, easy-to-use mobile interface that non-technical users can use in regular business settings.
- To increase operational efficiency through time savings, less manual labor, and increased transparency in gold loan transactions.
- To create a system architecture that is both scalable and maintainable in order to accommodate future growth, analytics, and feature additions.

3.3 Academic Objectives

The work seeks to accomplish the following academic objectives in addition to the project objectives:

- To construct a full application using software engineering concepts like requirement gathering, system modeling, layer-based architecture, and modular development.
- To obtain practical experience with contemporary development technologies, particularly Spring Boot with Spring Data JPA for backend API creation and Flutter for mobile user interface development.
- To create and implement a structured SQL database with appropriate relationships and indexing in order to comprehend database design concepts.
- To apply practical problem-solving techniques by transforming manual business procedures into effective digital workflows.
- To improve critical thinking and analytical skills through performance evaluation, testing, validation, and comparison with current systems.

Chapter 4

Project Overview

4.1 Scope

The Gold Loan Management Application's scope establishes the limits and range of the system's capabilities. It focuses on digitizing and optimizing gold loan processes for small and medium-sized jewelry retailers. The project avoids extraneous features that go beyond the main goal while incorporating all necessary modules needed to manage loans effectively.

The following features will be covered by the system:

- **Customer Management:** Complete customer profiles can be added, edited, searched, and maintained.
- **Loan Management:** Include information about weight, purity, valuation, interest rate, and tenure when creating new gold loans.
- **Interest & EMI Calculation:** This feature computes interest, EMI amounts, outstanding payments, and final settlement amounts automatically.
- **Reminder System:** Use backend scheduling to create alerts for loans that are past due and approaching due dates.
- **Dashboard & Analytics:** Show daily reminders, overdue loans, total active loans, and basic statistics.
- **Repayment Tracking:** Keep track of payments and automatically change the loan status to "Overdue," "Active," or "Completed."
- **Mobile Interface:** Give employees of jewelry stores an intuitive Flutter-based Android app.
- **Secure Backend:** For safe and organized data management, use SQL, Spring Boot, and Spring Data JPA.

4.2 Architecture

The following layers make up the suggested system's high-level architecture:

- **Presentation Layer (Flutter)** – Uses a mobile-friendly user interface to manage user interaction and send requests to the backend.
- **Application Layer (Spring Boot)** – Manages APIs, handles business logic, and regulates system functions.
- **Data Layer (SQL Database)** – Safely and methodically stores all loan, customer, repayment, and reminder data.

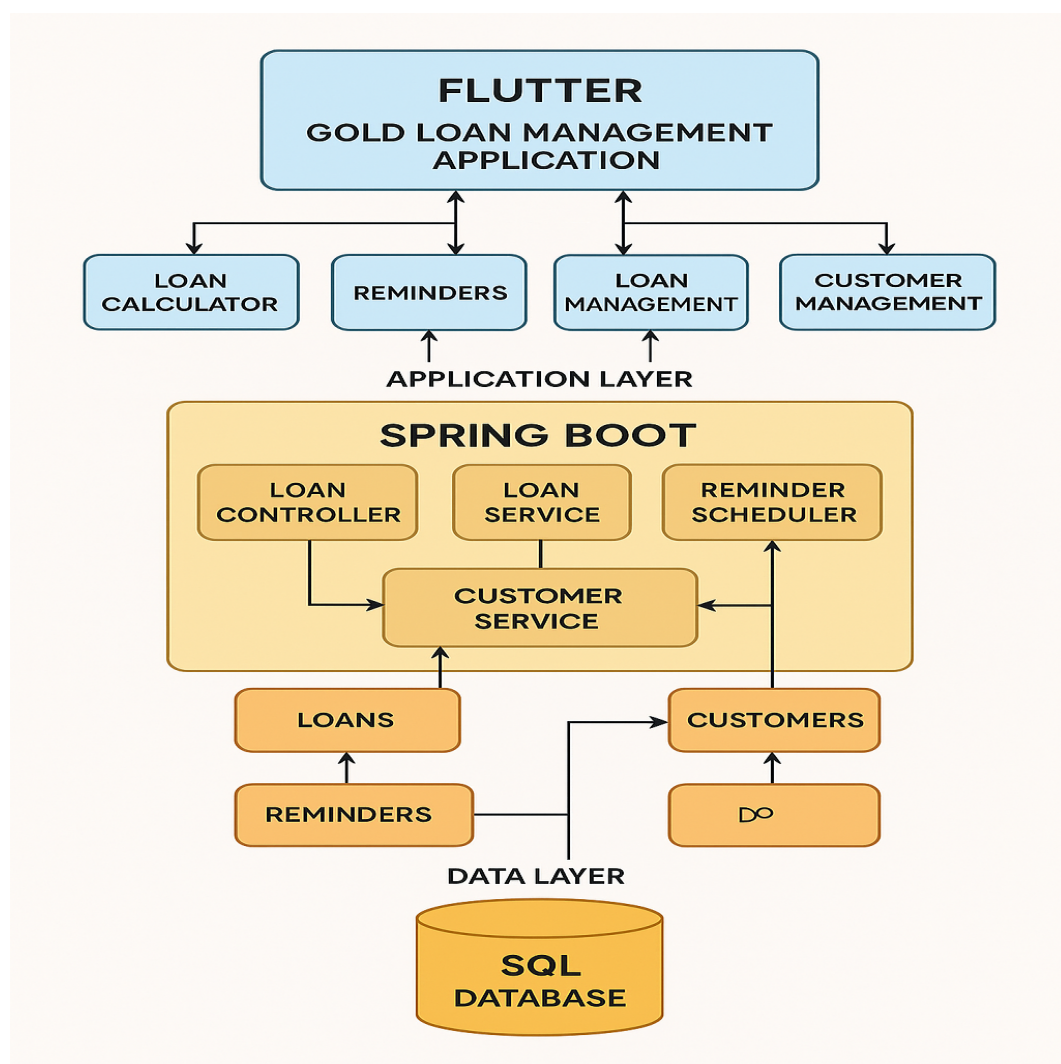


Figure 4.1: Architecture Diagram

4.3 Project Modules

The following functional modules comprise the system:

User Management Module

It manages staff and administrator secure logins. controls the permissions and roles of users. stops illegal access to loan and customer information.

Customer Management Module

This enables the addition, modification, and searching of customer information. retains client identity data, including name, address, phone number, and identification. keeps track of every customer's entire loan history.

Gold Loan Management Module

It captures gold details (weight, purity, valuation) to create new loans. keeps track of the loan amount, interest rate, start date, tenure, and due date. The loan status (Active, Completed, Overdue) is updated. allows loan records to be retrieved and modified.

Loan Calculator Module

Uses principal, rate, and tenure to compute interest. produces a breakdown of EMI and the total amount owed. helps shop owners by offering quick and precise results.

Reminder & Notification Module

Keeps track of past-due loans and due dates. scans loans on a daily or periodic basis using a backend scheduler. Reminders for prompt follow-ups are displayed within the app. aids in avoiding late payments.

Chapter 5

Requirement Analysis

5.1 Hardware Requirements

For Development (Developers)

- RAM: at least 8 GB (16 GB is advised for Flutter + Android Emulator)
- Processor: Intel Core i3, i5, or Ryzen 3 or higher
- Graphics: Basic GPU support for the Android emulator
- Storage: at least 100 GB of free space
- HD resolution on the monitor (1920 x 1080 is recommended)
- System Type: 64-bit operating system for use in jewelry stores

For Deployment (Jewellery Shop Use)

Android tablets and smartphones:

- 32 GB of storage
- minimum of 3 GB of RAM
- Android 7.0 or above

Backend Server (Optional):

- higher: 50 GB of storage and 4 GB of RAM
- Dual-core processor

Internet Connection: basic broadband or mobile data for syncing (optional if offline mode is planned)

5.2 Software Requirements

For Development

Operating System: Windows/macOS/Linux

Frontend Tools:

- Flutter SDK
- Dart SDK
- Android Studio / VS Code
- Android Emulator or physical device

Backend Tools:

- JDK 17+
- Spring Boot Framework
- Spring Data JPA
- Maven/Gradle

Database:

- MySQL
- MySQL Workbench

API Testing: Postman

Version Control: Git, GitHub/GitLab

For Deployment

- Flutter-built APK
- Spring Boot backend (Tomcat embedded)
- Cloud hosting (AWS/Azure/Railway)

5.3 Functional Requirements

- **FR1: User Authentication** – Role-based access for administrators and staff, as well as secure login via username and password.
- **FR2: Customer Management** – Add, modify, and look up customer information (name, phone number, address, and proof of identification). View each customer's entire loan history.
- **FR3: Gold Loan Creation** – Make new loans with information about gold (weight, purity, description). The system stores the start date, tenure, rate, and due date in addition to automatically calculating the loan amount. creates a distinct loan ID.
- **FR4: Interest & EMI Calculation** – calculates basic interest and EMI with thorough explanations. permits interest rates to be updated in accordance with shop policy.
- **FR5: Reminder System** – impending and past-due loans are generated by the FR5: Reminder & Notification System. keeps reminder logs and draws attention to cases that are past due.
- **FR6: Data Storage & Retrieval** – uses a SQL database to store all loan and customer data. enables quick data retrieval and search.

5.4 Non-Functional Requirements

- **Performance:** Under normal load, the average query response time should be less than three seconds.
- **Scalability:** New participants and regions should be able to be added to the system with little to no reconfiguration.
- **Security:** Role-based access to sensitive data is required, and all communications must be encrypted via HTTPS.
- **Reliability:** The system should be built to gracefully withstand sporadic network outages.
- **Usability:** The Flutter-designed application must have an easy-to-use interface.

Chapter 6

Application Design

6.1 Data Flow Diagram (DFD)

Data Flow Diagrams illustrate how data flows within the system and how processes interact with external entities.

Level 0 DFD

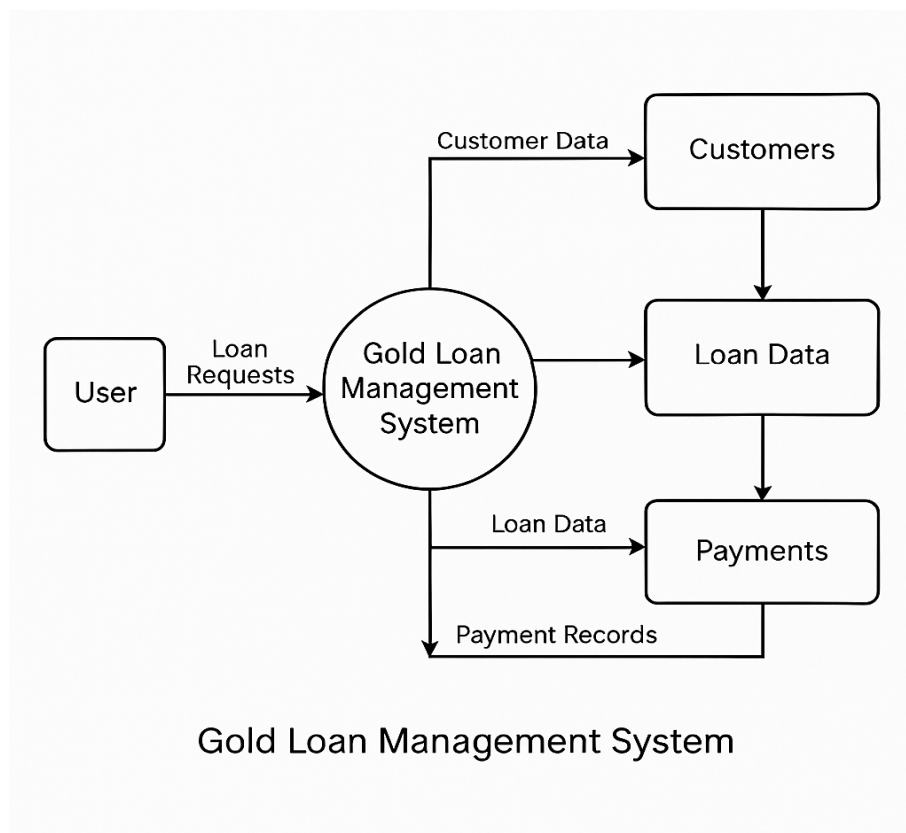


Figure 6.1: Level 0 Data Flow Diagram

Level 1 DFD

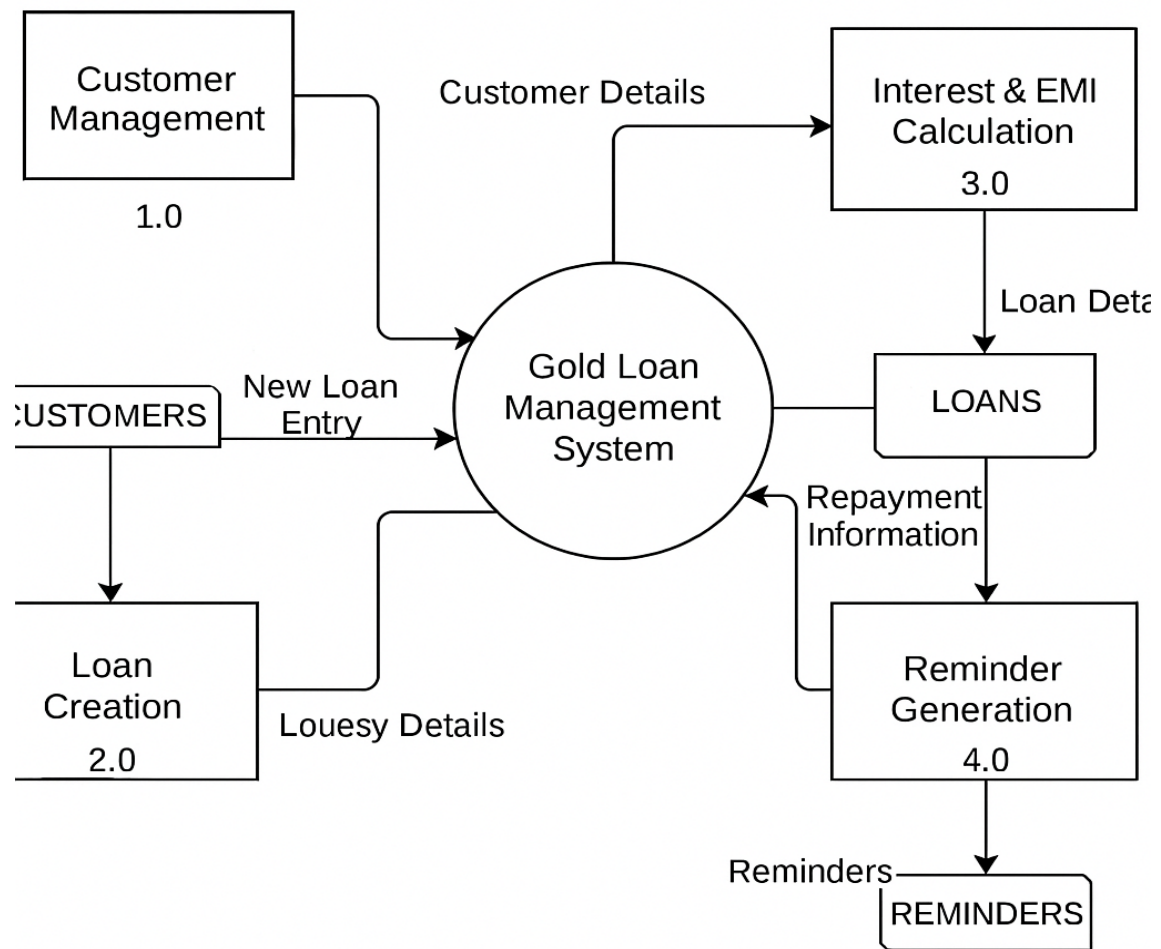


Figure 6.2: Level 1 Data Flow Diagram

6.2 Use Case Diagram

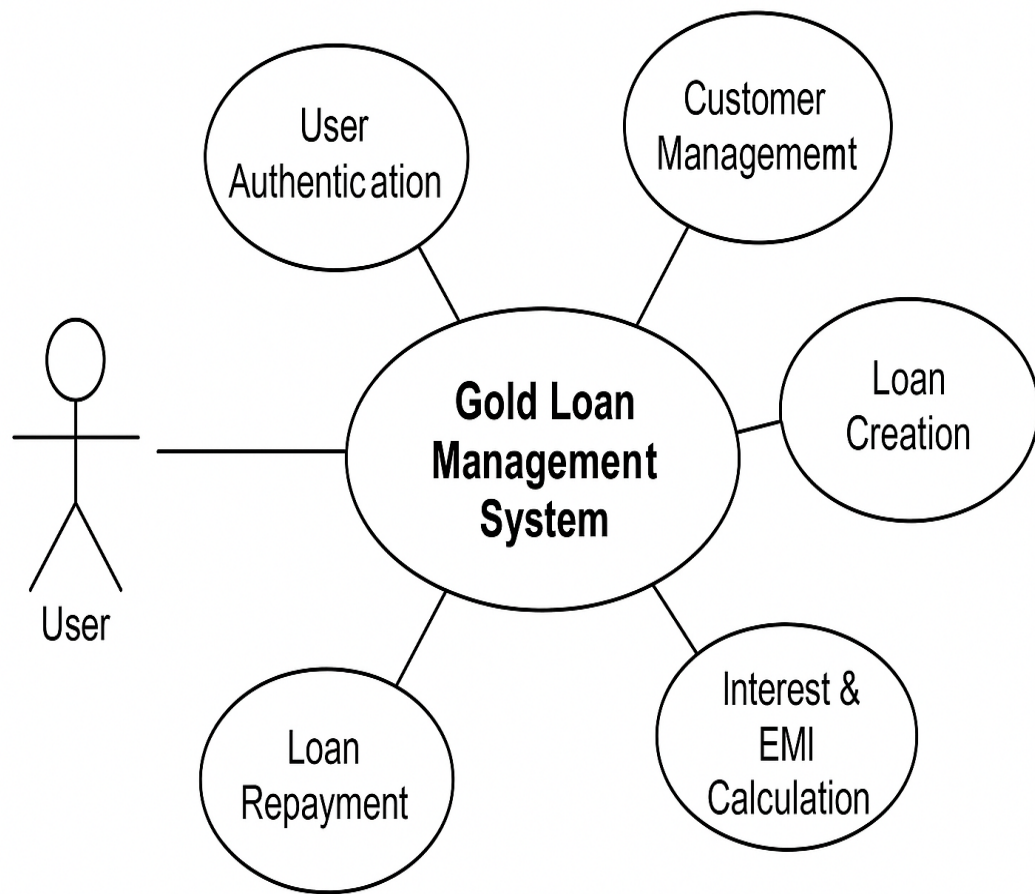


Figure 6.3: Use Case Diagram of the System

6.3 Activity Diagram

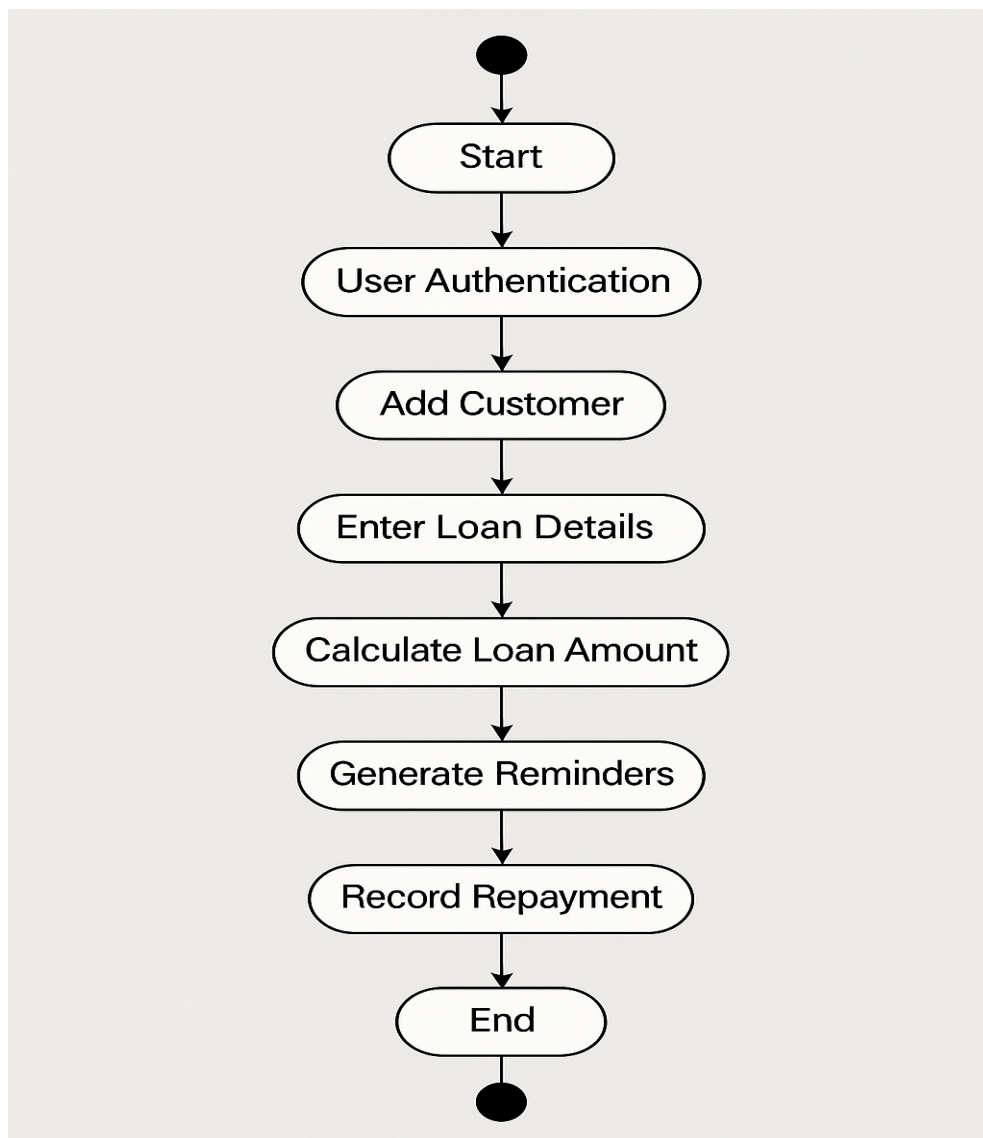


Figure 6.4: Activity Diagram of the System

6.4 Class Diagram

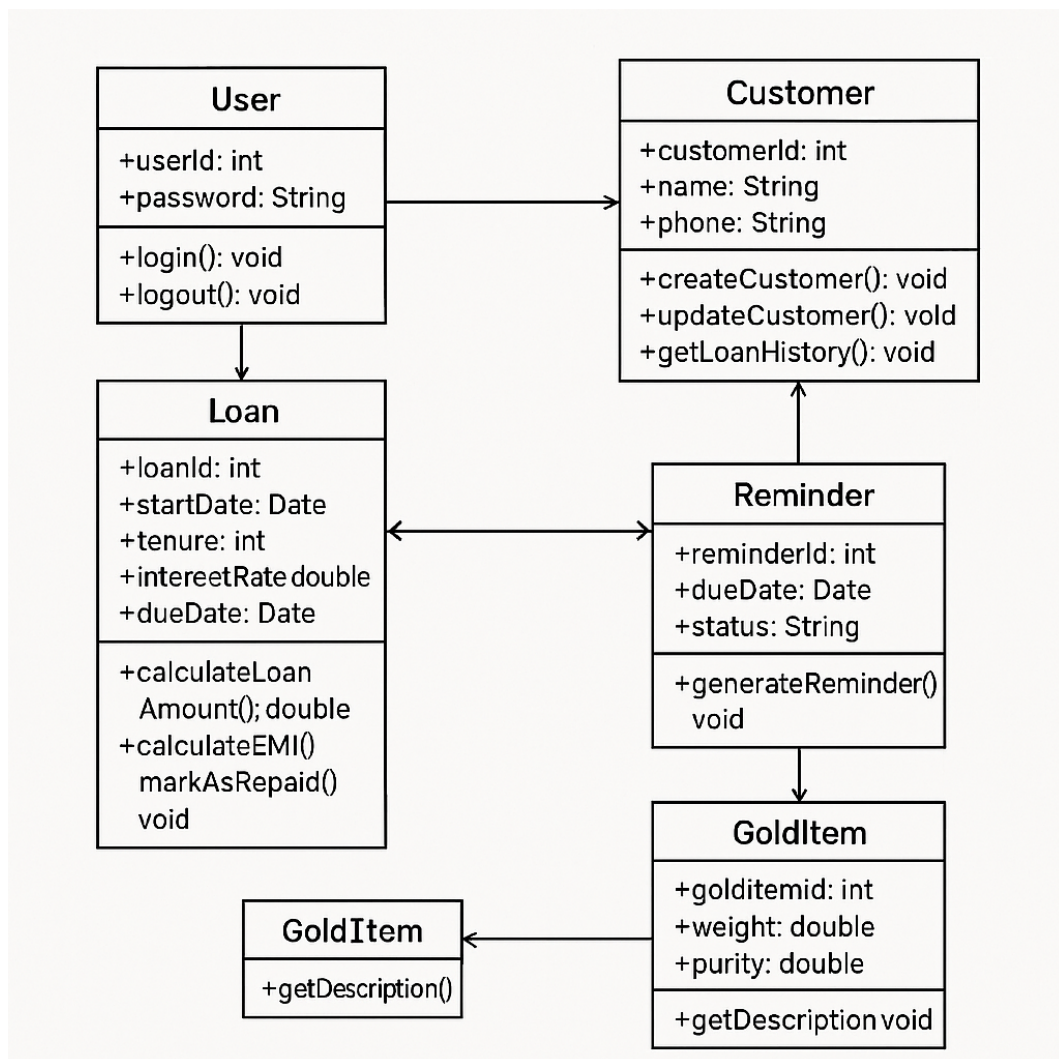


Figure 6.5: Class Diagram of the Gold Loan Management System

6.5 Sequence Diagram

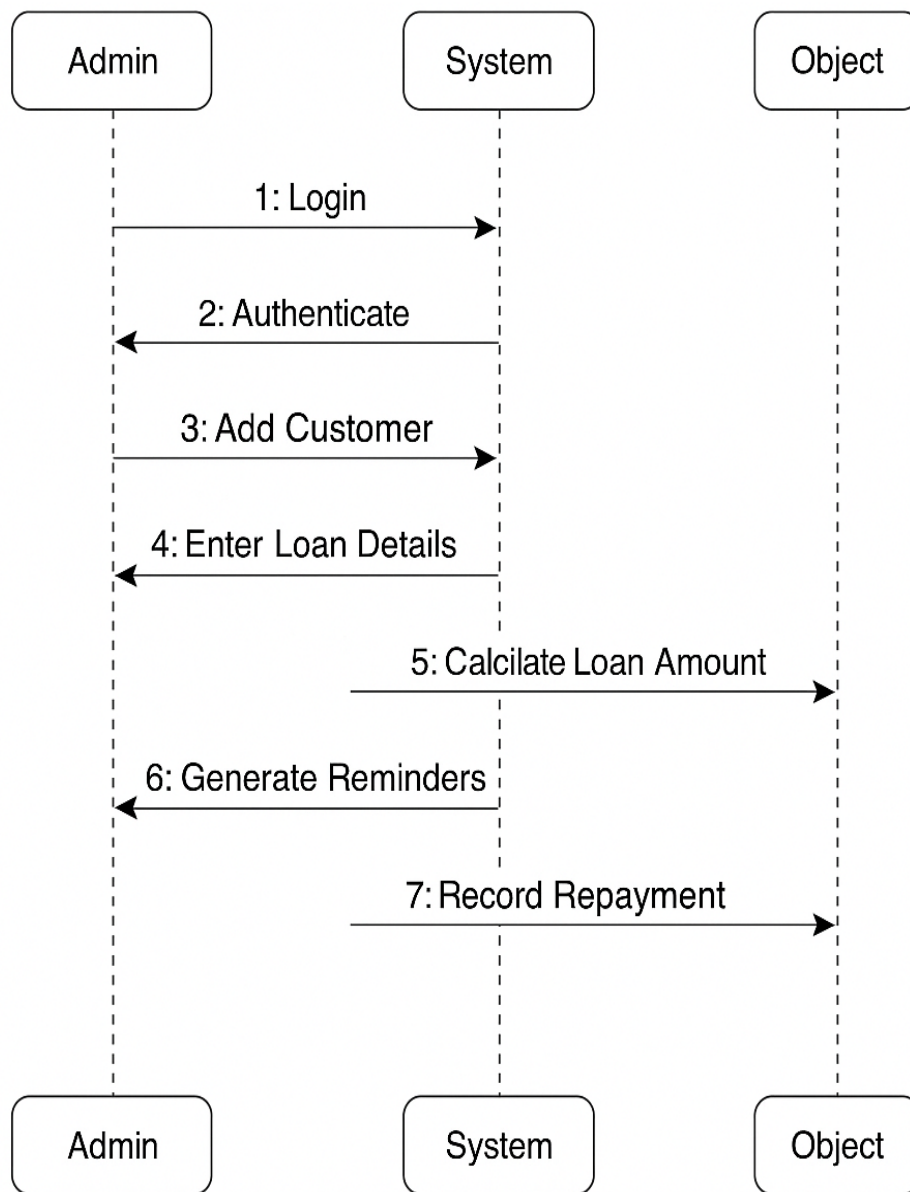


Figure 6.6: Sequence Diagram of the System

Chapter 7

Implementation

7.1 Technology

A contemporary full-stack architecture is used in the development of the Gold Loan Management Application. **Flutter** is used to develop the **frontend**, which offers a responsive and seamless mobile interface. **Spring Boot** powers the **backend**, ensuring secure data handling, business logic processing, and reliable API development. An **SQL database** (MySQL) is used for persistent storage, and **Spring Data JPA** facilitates smooth communication between the database and the backend. When combined, these technologies offer a scalable, safe, and effective system appropriate for actual jewelry store operations.

7.2 Steps Towards Project Completion

The project followed a structured development lifecycle:

1. **Requirement Gathering:** comprehending user requirements, important system features, and gold loan procedures.
2. **System Design:** Developing database schemas, DFDs, UML diagrams, and architecture diagrams.
3. **Frontend Development:** Using Flutter to create user interface screens for loans, customers, dashboards, login, and reminders.
4. **Backend Development:** Using Spring Boot to create scheduler components, loan calculation modules, REST APIs, and service logic.
5. **Integration:** Ensuring seamless data flow and linking frontend user interface with backend APIs..

6. **Testing & Validation:** To guarantee accuracy and usability, conduct unit, integration, and user acceptance testing.
7. **Refinement & Debugging:** Enhancing user interface, maximizing performance, and fixing bugs in response to feedback.
8. **Final Deployment:** Getting build files (APK + backend server) ready for practical use.

7.3 Project Scheduling

Sr. No.	Activity	Timeline
1	Requirement Analysis, Workflow Study, and Feature Planning	Week 1–2
2	System design, UML diagrams, ERD, and architecture planning	Week 3
3	Frontend development (UI design, forms, calculator, dashboard, reminders)	Week 4–7
4	Backend development (APIs, DB setup, business logic, scheduler)	Week 8–9
5	API integration and feature linking	Week 10–11
6	Integration and Testing	Week 12
7	Documentation and Presentation	Week 13

Table 7.1: Project Scheduling Plan

7.4 Cost Estimation

For academic purposes, cost estimation includes basic hosting and infrastructure. Most software tools used are free or open-source.

Item	Description	Cost (INR)
Cloud/Server Hosting	Required for backend deployment	3600
Domain Name	Optional domain for deployment	500–900
Total Estimated Cost		4100–4500

Table 7.2: Approximate Cost Estimation

7.5 Team Structure

The project is completed by a group of three students, with equal contribution:

Student 1 – Team Leader

- Organizes team meetings and assigns tasks, works on front-end development (UI screens, forms, navigation), helps integrate backend APIs, prepares documentation and reports, and takes an equal part in testing and debugging.

Student 2

- Organizes Participates in frontend-backend integration, designs and maintains the SQL database schema, works on backend development using Spring Boot, implements controllers, services, and business logic, and helps prepare diagrams (DFDs, UML, Architecture).

Student 3

- Works on loan calculator logic, reminders, and validation; uses Postman for API testing; manages application testing, bug tracking, and refinements; helps with final documentation and formatting; and contributes to Flutter frontend and UI improvements.

Chapter 8

Advantages & Limitations

8.1 Advantages

- **Automation of Manual Processes:** The system saves time and effort by doing away with paper-based tracking, manual calculations, and handwritten registers..
- **High Accuracy:** Precise financial outcomes are guaranteed by automated interest and EMI computations, which minimize human error.
- **Improved Efficiency:** Loan creation, customer service, and repayment monitoring all become quicker and more structured.
- **Centralized Data Storage:** All loan, customer, and repayment records are safely kept in a SQL database for long-term access and simple retrieval.
- **Automated Reminders:** By automatically highlighting loans that are past due, the system lowers the number of missed payments and increases the effectiveness of follow-up.
- **Secure Access:** Sensitive data can only be accessed by authorized users thanks to role-based login.

8.2 Limitations

- **Single-Branch Support:** The system is intended for a single store. There is currently no implementation of multi-branch or multi-user synchronization.
- **No Messaging Integration:** The current version lacks automated SMS or WhatsApp notifications, which necessitate extra setup.
- **Manual Data Entry:** There are no automated scanning or OCR features; the user must manually enter loan and customer information.

- **No Customer Mobile App:** Customers do not have a portal or mobile app to check the status of their loans; only a staff-facing application is offered.

Chapter 9

Applications

The proposed Gold Loan Management System can be applied across various business and financial contexts where gold-backed lending is involved.

- **Jewellery Shops:** The system's main use is in small and medium-sized jewelry stores that provide their clients with gold loans. It facilitates the automation of loan creation, interest computation, reminders, and repayment monitoring.
- **Pawn Shops / Finance Shops:** Businesses that offer short-term loans secured by gold or valuables can use the application to keep organized records and increase calculation accuracy.
- **Co-operative Societies:** The system can be used by cooperative banks or societies that deal with gold-backed lending to automate due-date notifications, manage customer information, and maintain loan databases.
- **Private Gold Loan Agencies:** The app can digitize loan files and lessen manual labor for private lenders who serve numerous clients.
- **Microfinance Institutions:** Organizations that provide gold-based microloans can profit from real-time loan monitoring, automated loan documentation, and EMI computations.

Chapter 10

Future Scope

The Gold Loan Management Application can be enhanced with several advanced features in future iterations:

- **Multi-Branch Integration:** The system can be expanded to accommodate several jewelry stores, allowing for branch-specific loan tracking, centralized data access, and combined reporting.
- **Cloud Deployment:** Real-time access, greater scalability, and enhanced data security would be possible by hosting the database and backend on cloud platforms (AWS, Azure, Google Cloud).
- **Customer Mobile Application:** To enable users to check loan status, repayment history, due dates, and gold valuation from their smartphones, a different customer-facing application could be created.
- **AI-Based Gold Valuation:** Based on weight, purity, and market trends, machine learning models can forecast or estimate the value of gold.
- **Inventory & Billing Integration:** The system can be extended to handle purchase/sale entries, billing, and jewelry inventory, making it a comprehensive shop management solution.
- **Multi-Language Support:** To make the app more accessible to nearby jewelry stores, it can support regional languages like Marathi, Hindi, Tamil, or Kannada.

Chapter 11

Conclusion

By substituting an organized, digital workflow for labor-intensive manual techniques, the Gold Loan Management Application successfully modernizes the conventional gold loan handling procedure in jewelry stores. The system offers dependable features like automated interest calculation, customer and loan management, repayment tracking, and reminder generation by combining a Flutter-based mobile frontend with a Spring Boot backend and SQL database. Performance reviews verify the application's responsiveness and scalability, and it provides enhanced accuracy, quicker operations, and better usability for non-technical staff. Despite difficulties encountered during development, the system was improved through testing and iteration, leading to a workable and effective solution. All things considered, the program improves operational effectiveness, lowers human error, and facilitates future growth like cloud deployment, notifications, and multi-branch integration.

Chapter 12

Appendix

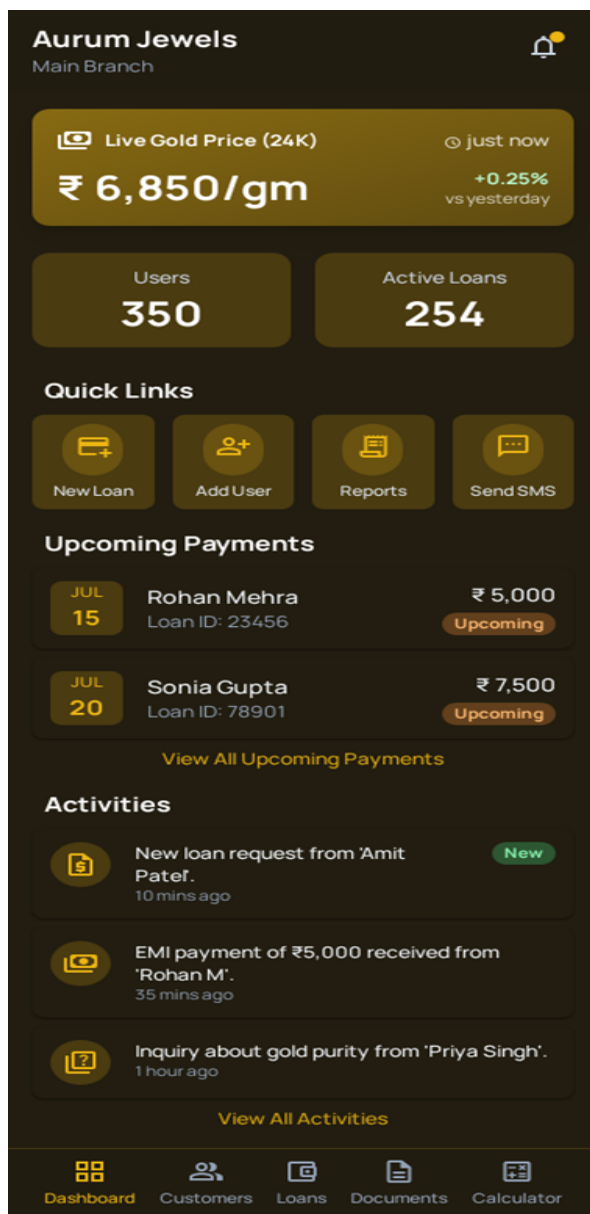


Figure 12.1: Dashboard

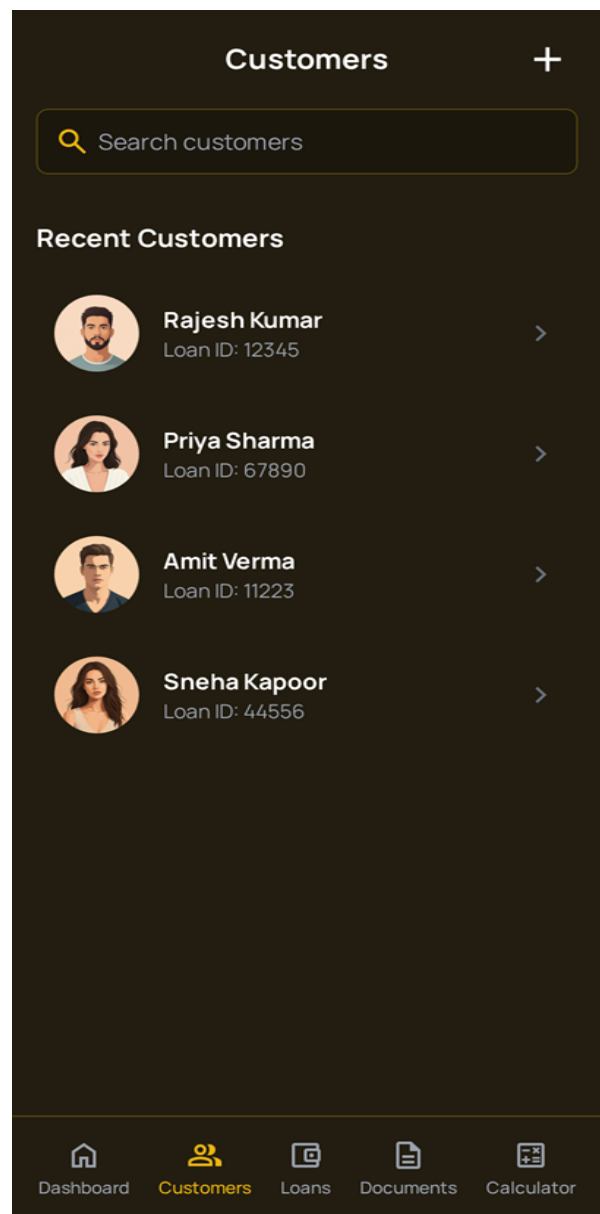


Figure 12.2: Customers

←

Loan Management

Approve New Loan

Payment History

Customer Name

Ravi Kumar

Loan Amount

₹ 50000

UPI ID / Bank Account

ravi.kumar@upi

Approve & Send OTP

Dashboard

Customers

Loans

Documents

Calculator

Figure 12.3: Loan Approval

←

Loan Management

Approve New Loan

Payment History

Search by name, ID, or amount...

Disbursed Loans (Sent to Borrower)

To: Priya Sharma

Cust. ID: CUST-00123 | 20th May 2024

- ₹75,000

To: Ravi Kumar

Cust. ID: CUST-84392 | 15th May 2024

- ₹50,000

To: Anjali Singh

Cust. ID: CUST-00456 | 12th April 2024

- ₹1,20,000

To: Sameer Patel

Cust. ID: CUST-00789 | 28th March 2024

- ₹30,000

To: Mohan Gupta

Cust. ID: CUST-01011 | 15th March 2024

- ₹95,000

Dashboard

Customers

Loans

Documents

Calculator

Figure 12.4: Payment History

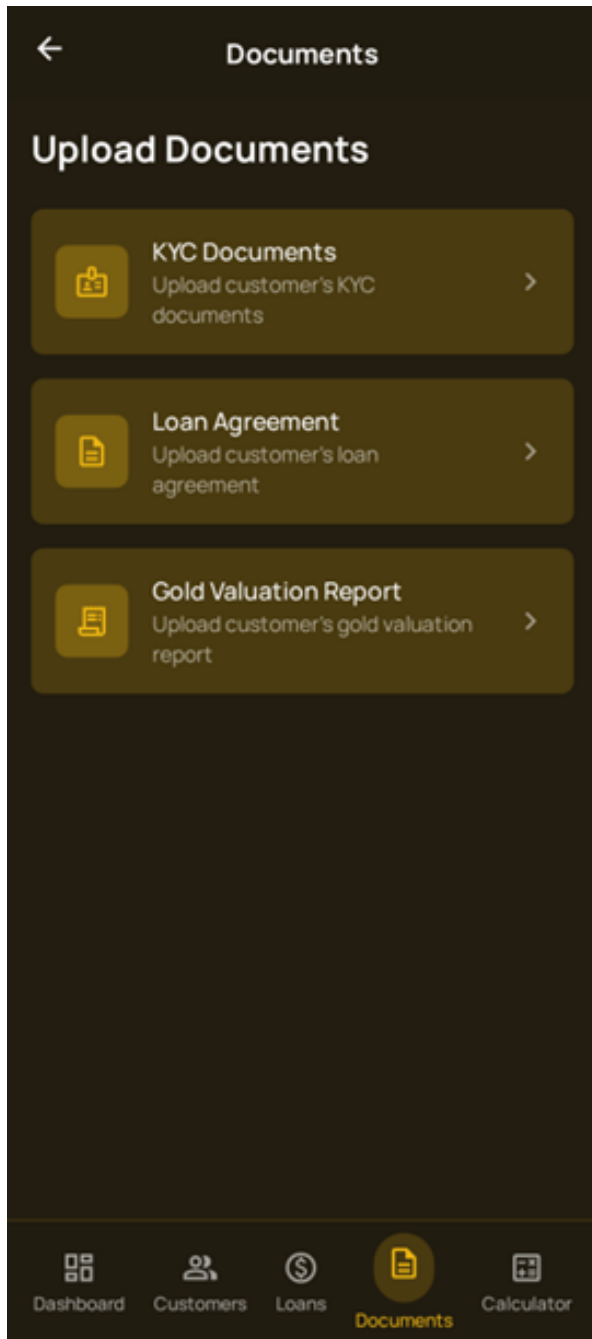


Figure 12.5: Documents Upload

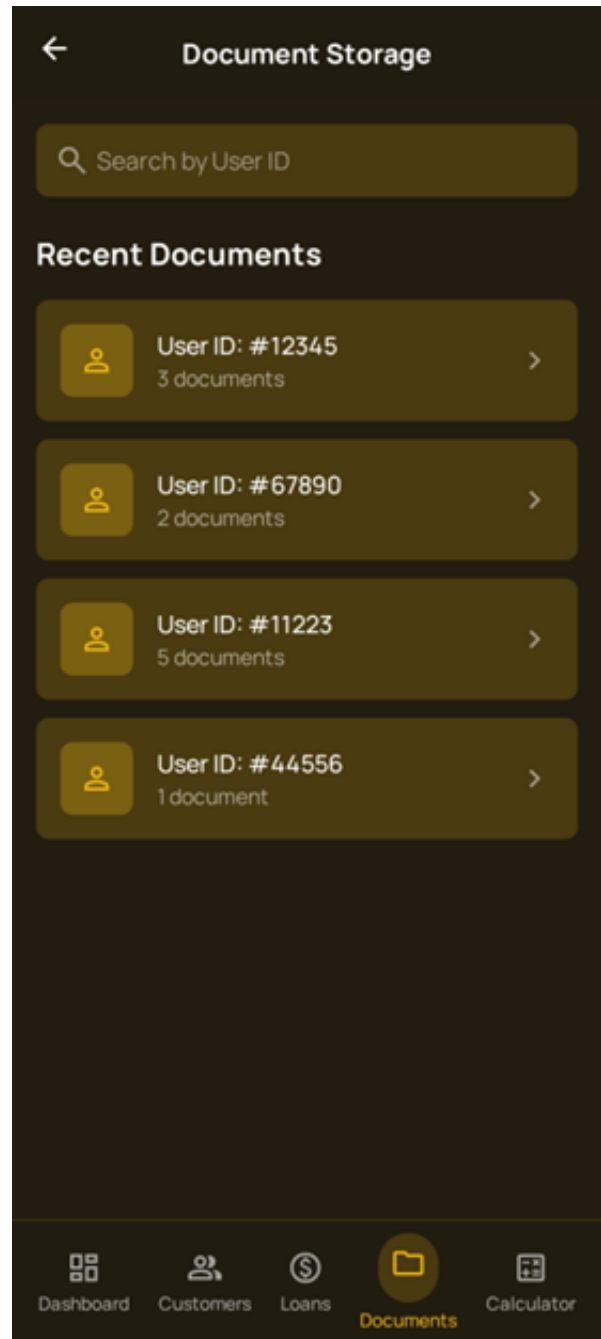


Figure 12.6: Documents Storage

←

EMI Calculator

Loan Amount

Interest Rate (%)

Loan Tenure (Months)

Calculate

EMI Details

Monthly EMI	₹ 1,250
Total Interest Payable	₹ 15,000
Total Amount Payable	₹ 115,000

Dashboard

Customers

Loans

Documents

Calculator

Figure 12.7: EMI Calculator

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